IIT Ropar

AI101 Mini Project

AQI Hazardous Prediction

Machine Learning-Based Classification of Hazardous Air Quality Days



Project Introduction

The project aims to develop a Machine Learning model that predicts whether a particular day has hazardous air quality based on pollutant measurements. Using air quality data and Logistic Regression, the model classifies days into Hazardous and Non-Hazardous categories, helping identify potential air pollution risks.

Team Members & Contributions

Vineet Shekhar
Documentation Lead

- Prepared project documentation
- Compiled reports and insights

Rishabh Vishnoi
Execution Lead

- Developed ML model
- Handled data preprocessing and model training

Sandeep Kumar
PPT Lead

- Designed presentation slides
- Prepared visual content & data explanation

Divya Godara
Video Lead

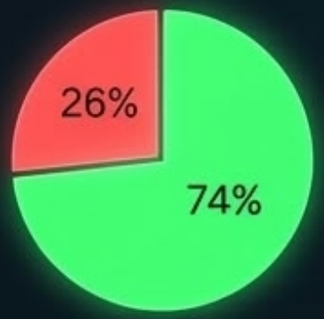
- Created project video
- Edited and compiled final video presentation

PREDICT • PREVENT • BREATHE BETTER

CELL 1 PREPROCESSING WORKFLOW

Vineet Shekhar | AQI Project

CLASS DISTRIBUTION



- 74% Safe (Green)
- Hazardous (Red)

Processed data: 19531 total

CLASS DISTRIBUTION



- 74% Safe (Green)
- 26% Hazardous (Red)

Processed data: 74%, 70%

Raw Data:
city_day.csv
29531 records

PM2.5 NO_2 NTC_2
ETC_s NO_z etc.

```
df.interpolate(eethod='eedian')
```

Median Missing Value Imputation

```
df.interpolate(method='median')
```

Target Labeling:
Hazardous (AQI>200) vs. Safe

80-20 Stratified Train-Test Split

PM2.5
NO_2
NO_2
NO.

Training Dataset:
19880 Shapes

19880

Testing Dataset:
4970 Shapes

4970

AQI HAZARDOUS PREDICTION PIPELINE

PROBLEM FRAMING • EXPLORATORY ANALYSIS • MODEL PERFORMANCE

Logistic Regression | city_day.csv

01 / PROBLEM FRAMING

Raw city-day records to a binary risk label

29,531	24,850	12
RAW RECORDS	CLEANED	FEATURES

- Target 0: Safe / Moderate (AQI ≤ 200)
- Target 1: Hazardous (AQI > 200)
- 80 / 20 stratified train–test split
- Median imputation; no missing values remain

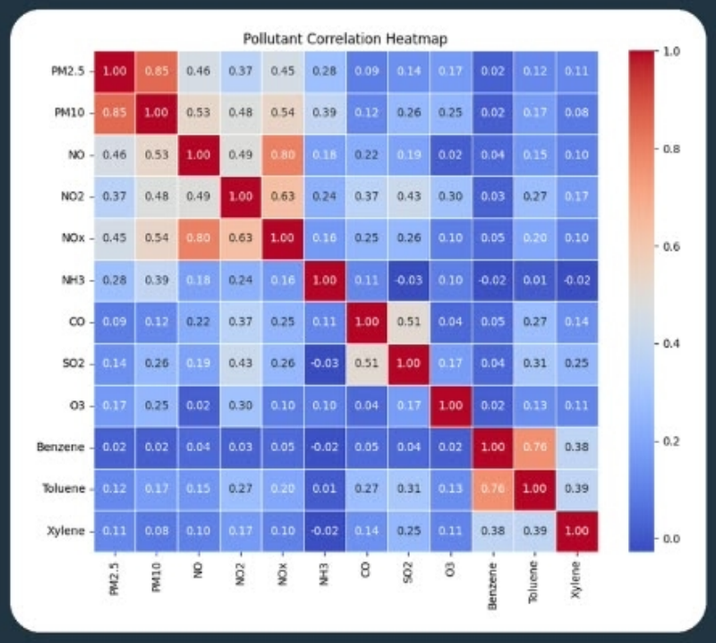


CLASS BALANCE

74.02%	25.98%
SAFE / MODERATE	HAZARDOUS

02 / EXPLORATORY DATA ANALYSIS

Pollutant relationships reveal strong collinearity



KEY CORRELATIONS

r = 0.85	r = 0.80	r = 0.76
PM2.5 ↔ PM10	NO ↔ NOx	Benzene ↔ Toluene

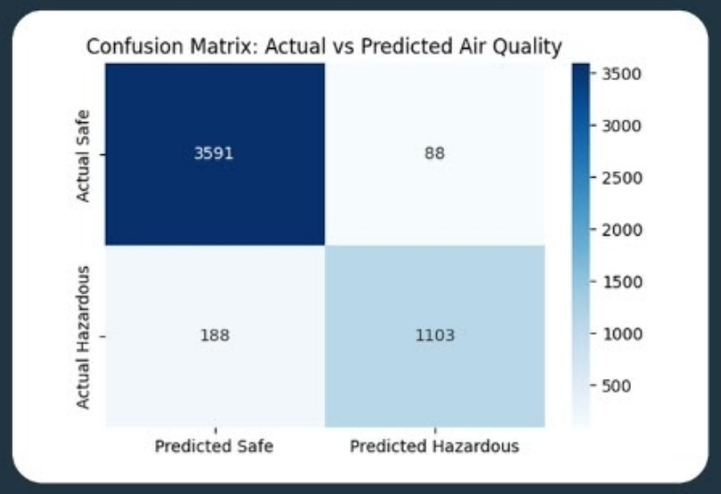
Highly correlated inputs carry overlapping signal.
Interpret individual feature effects with that context.

03 / MODEL PERFORMANCE

Logistic Regression performs consistently on held-out data

94.45%	0.8888	0.1486
TEST ACCURACY	TEST F1-SCORE	TEST LOG LOSS

94.36%	0.8871
TRAIN ACCURACY	TRAIN F1



TEST SET • 4,970 DAYS

3,591	1,103
TRUE SAFE	TRUE HAZARDOUS

Team: Vineet Shekhar • Rishabh Vishnoi • Divya Godara • Kollati Sandeep Kumar

Train vs test accuracy differ by only 0.09 percentage points

ADVANCED AIR QUALITY CLASSIFICATION MODEL EVALUATION

CELL 4: ADVANCED MODEL EVALUATION DASHBOARD | ADVANCED MODELING & CLASSIFICATION ANALYSIS

PROJECT CONTEXT:
Project Component: Section 4 - Model Optimization, Threshold Tuning & Risk Classification

AUTHOR:
DIVYA GODARA | VIDEO LEAD & COMPONENT OWNER

--- FINAL CLASSIFICATION REPORT ---

	precision	recall	f1-score	support
Safe (0)	0.95	0.98	0.96	3679
Hazardous (1)	0.93	0.85	0.85	1291
accuracy			0.94	4970
macro avg	0.94	0.92	0.93	4970
weighted avg	0.94	0.94	0.94	4970

Key Metrics Summary

ACCURACY: **94%**



HAZARDOUS RECALL: **85%**



HAZARDOUS F1-SCORE: **89%**



CONFUSION MATRIX: ACTUAL VS. PREDICTED AIR QUALITY

True Safe (3591): Correctly predicted non-hazardous

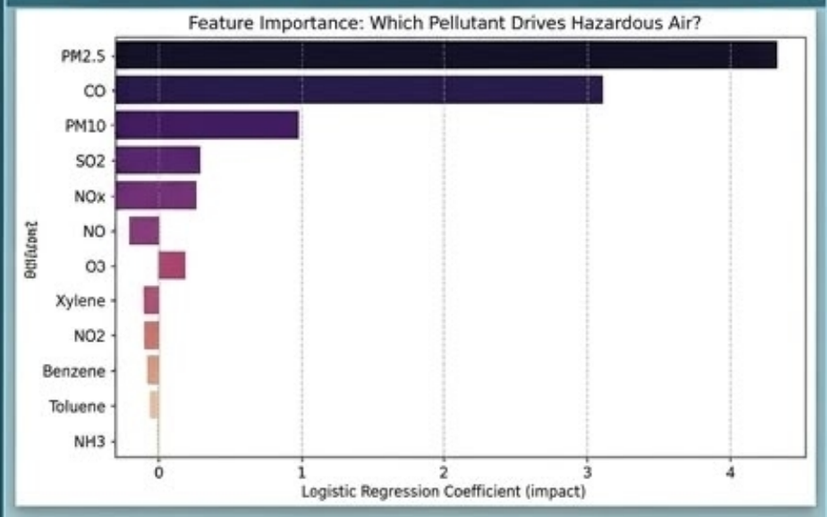
True Hazardous (1103): Correctly identified risk days



False Safe (188) (CRITICAL MISSES): Missed Hazardous Events (Recall focus)

False Hazardous (88): False Alarms

FEATURE IMPORTANCE: WHICH POLLUTANT DRIVES HAZARDOUS AIR?



- **PM2.5:** Primary driver (<4 coeff.)
- **CO:** Significant impact (>3 coeff.)
- **PM10:** Notable contributor (~1 coeff.)

ADVANCED MODELING OBJECTIVES MET

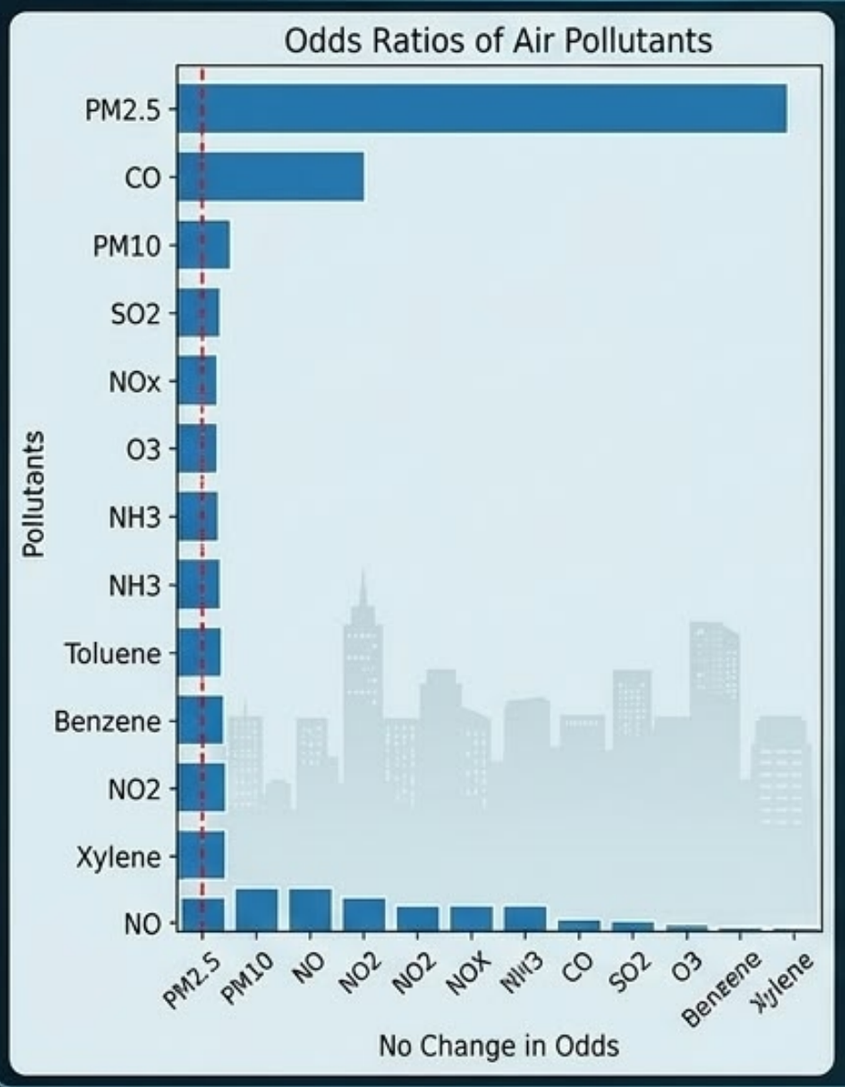
- Optimized precision, recall, & F1 for spikes (Precision Focus)
- Threshold tuning optimized sensitivity for borderline events (Recall Focus)
- GridSearchCv for advanced classifiers

ADVANCED AIR QUALITY MODEL ANALYSIS

SECTION 5: ADVANCED MODEL ANALYSIS

AUTHOR: KOLLATI SANDEEP KUMAR

I. ODDS RATIO ANALYSIS



KEY DRIVERS

PM2.5 increases hazardous odds

73.53x

per std-dev

CO increases odds

22.07x

Odds Ratios

(Impact per 1 Std-Dev Increase)

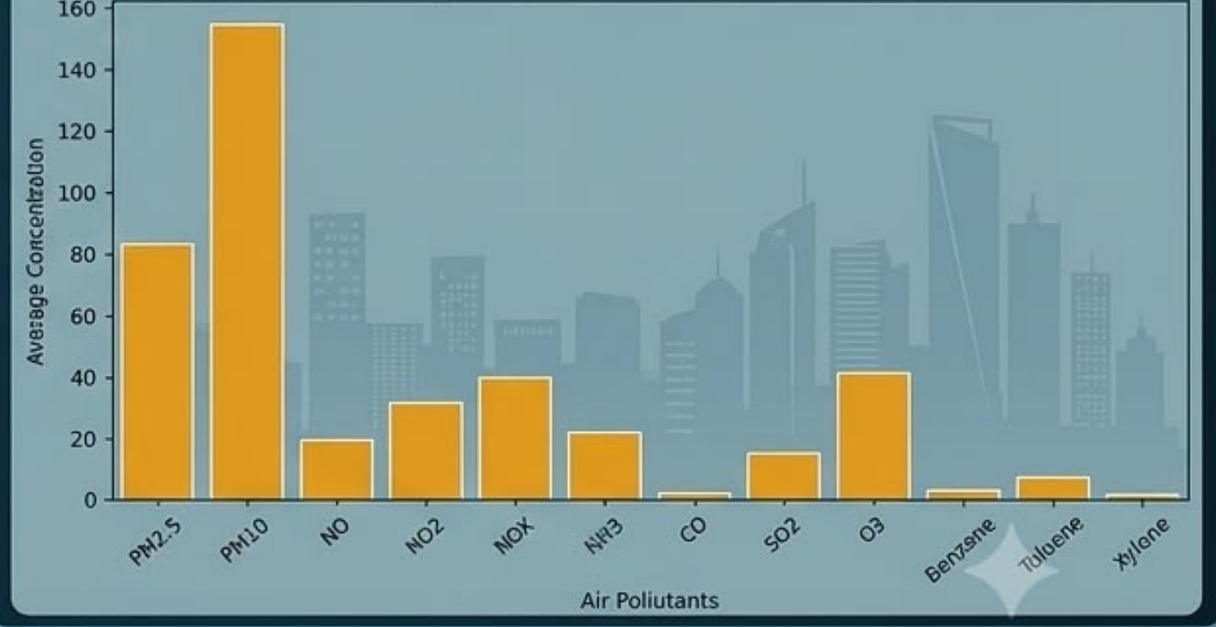
PM2.5	(73.5339)
CO	(22.0709)
PM10	(2.6723)

II. TIPPING POINT ANALYSIS

174

Borderline Days Identified

Days with 40% – 60% probability of being Hazardous



=== ADVANCED ANALYSIS COMPLETED ===

✓ Odds Ratios calculated ✓ Borderline days identified ✓ Average levels calculated

Contributor: Kollati Sandeep Kumar